Business Requirement Document (BRD)
Smart Insurance Claim Management System

1. Business Objective

The objective of this project is to build a scalable microservice-based insurance claim management platform to manage customers, policies, claims, and notifications.

2. Business Problems

Current insurance operations are manual, claim processing is slow, notifications are delayed, and configuration changes require application restarts.

3. Proposed Solution

Build a distributed microservice ecosystem using Spring Boot, Spring Cloud Config, RabbitMQ, Docker, and H2 databases.

4. Functional Modules

Customer Service: Manage customer onboarding and updates.
Policy Service: Manage insurance policies and premium calculations.
Claim Service: Raise and approve insurance claims.
Notification Service: Send notifications asynchronously.

5. Business Rules

• Customer email must be unique.
• Customer age must be >= 18.
• Policy amount must be > 50,000.
• Claim amount cannot exceed policy amount.
• Claims above 10 lakhs require manual approval.

6. Claim Workflow

OPEN → UNDER_REVIEW → APPROVED/REJECTED → SETTLED

7. Runtime Configuration Requirements

Configuration should be managed centrally using Spring Cloud Config Server. GitHub webhooks and Spring Cloud Bus should refresh configurations dynamically without restarting services.

8. Notification Requirements

Notification service should process asynchronous messages using RabbitMQ to ensure loose coupling and retry capabilities.

9. Deployment Requirements

All microservices must be containerized using Docker and orchestrated using Docker Compose.

10. Non-Functional Requirements

• Scalability
• Fault Tolerance
• Maintainability
• Runtime Configuration Refresh
• Audit Logging